

Kunal Pramanik

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Education

Dhirubhai Ambani University (DA-IICT)

Master of Science – Data Science (M.Sc.)

- Relevant Skills: Python, PostgreSQL, Machine Learning

Gandhinagar, Gujarat, India

Jul 2025 – Present

Scottish Church College

Bachelor of Science – Mathematics (B.Sc.) | **Grade: A**

- Relevant Skills: Mathematical concepts, C Programming, Statistical Analysis

Kolkata, West Bengal, India

Jul 2021 – Jul 2024

Projects

SarkariYojana AI | Groq LLaMA 3, RAG, Supabase, Firebase | 🐙 github.com/Kunal-Pramanik/Sarkari_yojana_ai

- Engineered a production-grade AI chatbot for Indian government scheme discovery, combining a FAISS-based RAG pipeline over 4,600+ schemes with real-time web search via Serper API.
- Architected a full-stack system with JWT-based authentication (Supabase Auth), persistent cross-session chat history (Firebase Firestore), and semantic long-term memory using vector embeddings.
- Implemented a topic-aware guardrail system that classifies off-topic queries and redirects users to relevant schemes, preventing LLM hallucination outside the knowledge domain.
- Built a streaming Next.js frontend with session-based sidebar history, markdown rendering, and source citation cards; deployed backend on Render and frontend on Vercel.
- Applied skills: LLM Engineering, RAG, Vector Search, Full-Stack Development, Cloud Deployment, Auth Systems.

IllumiRefine AI | PyTorch, OpenCV, FastAPI, React, Hugging Face | 🐙 github.com/Kunal-Pramanik/IllumiRefine-AI

- Architected a full-stack computer vision platform for extreme low-light image enhancement utilizing a highly efficient Self-Calibrated Illumination (SCI) neural network.
- Engineered a novel “SCI-Guided Adaptive Fusion” post-processing pipeline, dynamically combining fractional gamma correction with an illumination-guided NLM denoising mask.
- Successfully eliminated severe sensor noise in pitch-black regions while strictly preserving High Dynamic Range (HDR) edge structures without overexposure.
- Deployed a React frontend on Vercel and a CPU-optimized PyTorch/FastAPI inference engine on Hugging Face, achieving sub-2-second end-to-end processing speeds.
- Applied skills: Deep Learning, Computer Vision, Image Processing, API Development, Cloud Deployment.

Connect2Faculty | Python, FastAPI, NLP, SQLite, Vercel V0, Render | 🐙 github.com/Kunal-Pramanik/Connect2Faculty

- Architected an end-to-end Semantic Search Engine for Faculty Discovery, associated with Dhirubhai Ambani University.
- Built a custom Big Data Engineering (BDE) pipeline using BS4 and FastAPI to scrape, clean, and index faculty data at scale.
- Implemented 384-dimensional vector embeddings using MiniLM for high-accuracy semantic similarity matching.
- Deployed a full-stack Next.js frontend on Vercel V0 with a FastAPI backend hosted on Render; leveraged SQLite for lightweight vector storage.
- Applied skills: Web Scraping, Data Engineering, NLP, Data Analysis, EDA, Machine Learning.

Bankruptcy_Predictor | Python, Random Forest, Streamlit, EDA | 🐙 github.com/Kunal-Pramanik/Bankruptcy_predictor

- Developed an interactive ML dashboard to predict company bankruptcy risk using real-world financial indicators.
- Engineered features and performed comprehensive Exploratory Data Analysis (EDA) to identify key financial risk signals.
- Trained and optimized a Random Forest classifier achieving high predictive accuracy; deployed via Streamlit for real-time inference.

Licenses & Certifications

IBM Internship – Data Analytics

IBM

Credential ID: PLAN-3C665A29A398

Oct 2023

- Completed IBM-certified internship program with focus on Data Analysis and Statistical Data Analysis.

Technical Skills

Languages & Databases: Python, C, SQL/PostgreSQL

Machine Learning & AI: Machine Learning, NLP, EDA, Statistical Data Analysis, Vector Embeddings

Frameworks & Cloud Tools: FastAPI, Streamlit, Vercel V0, Render, Linux

Data Engineering: Web Scraping , Data Engineering Pipelines, Data Analysis

Other: Git, GitHub, IBM Data Analytics (Certified)

Leadership & Volunteering

President – Wann Hostel

Scottish Church College

Student Leadership & Administration

Jul 2022 – Jun 2024

- Elected to lead and represent the student body of Wann Hostel, managing daily operations and student welfare for a diverse community.
- Coordinated hostel events and acted as a primary liaison between students and college administration to resolve infrastructure and residential issues.

Mathematics Educator

Sabujmon School, Scottish Church College

Volunteer Educator – Children (Underprivileged)

May 2022 – Jun 2024 (2 yrs 2 mos)

- Served as a Mathematics Educator at Sabujmon School, an adopted school for underprivileged children, for over 2 years.
- Designed and delivered engaging math lessons, improving foundational numeracy skills in a resource-constrained environment.